Centered Natural-Parameter Group Selection for Posterior-Score Contrast Support in von Mises–Fisher Mixtures

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Abstract

Finite mixtures of von Mises–Fisher distributions are used for directional clustering, but existing sparse formulations target component prototypes rather than coordinates that separate posterior scores. We define posterior-score contrast support using centered natural parameters and propose the centered natural-parameter group lasso (E-CGL), which penalizes coordinatewise contrasts while leaving common effects unpenalized. Estimation combines a guarded proximal generalized-EM path, support-constrained refitting, and information-criterion selection conditional on the number of components. We establish a decomposition of canonical heterogeneity, label-invariant support, existence on a bounded concentration space, the closed-form proximal map, and conditional convergence of the fixed-responsibility subproblem. For an idealized inexact generalized-EM sequence, we establish a stationary-accumulation-point result under regularity and vanishing-residual conditions; global optimality is not claimed. Oracle-error-calibrated simulations show improving sparse-support recovery with sample size and expose limitations under dense or weak support. In Classic3 after cleaning (n = 3,883, d = 2,000), E-CGL selected 1,421 coordinates and attained an adjusted Rand index (ARI) of 0.9761 and normalized mutual information (NMI) of 0.9539, compared with ARI 0.9760 and NMI 0.9538 for the dense component-specific-concentration fit. Five-split E-CGL analyses gave a mean conditional support Jaccard index of 0.933 in Classic3, whereas BBCSport

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showed held-out density loss under sparsification. E-CGL is intended for sparse or compressible posterior-score contrasts, not as a general replacement for dense or prototype-oriented mixtures.

Keywords: Directional data; mixture model; von Mises-Fisher distribution; heterogeneity selection; group lasso; posterior-score contrast support

1. Introduction

Directional representations arise whenever only the orientation of an observation is meaningful. Common examples include normalized document vectors, compositional profiles after normalization, and embedding vectors on the unit hypersphere. The von Mises–Fisher (vMF) distribution is a standard model for such data, and finite vMF mixtures provide a model-based alternative to spherical k-means (Banerjee et al., 2005). In high dimensions, however, the fitted component structure can involve many coordinates, making both estimation and interpretation difficult.

Sparse vMF mixtures have primarily been formulated through sparsity of the component direction vectors. Rossi and Barbaro (2022), for example, penalize their entries to estimate sparse directional prototypes. This target is appropriate when the scientific question concerns which coordinates describe each component prototype. It is not identical to asking which coordinates distinguish components in posterior classification. A coordinate can be active in every prototype and still contribute almost equally to all component scores. Conversely, a coordinate with a modest directional loading can be influential when component concentrations differ. The distinction is therefore between prototype support and the feature-dependent linear support of posterior component comparisons.

The construction draws on contrast penalties and grouped variable selection, which address related but distinct estimands. Penalized Gaussian mixtures have used entrywise mean shrinkage, grouped component means, and pairwise mean fusion (Pan and Shen, 2007; Xie et al., 2008; Guo et al., 2010). Sum-to-zero factor contrasts have also been used to separate common levels from heterogeneity (Bondell and Reich, 2009), while other directional-mixture methods target feature saliency, block structure, or hierarchical parameter regularization (Amayri and Bouguila, 2013; Salah et al., 2016; Gopal and Yang, 2014; Taghia et al., 2014; Salah and Nadif, 2019). Discriminative subspace approaches provide another route to variable relevance in model-based clustering (Bouveyron and Brunet, 2012; Bouveyron and Brunet-Saumard, 2014). The remaining gap is to connect equality-based coordinate selection to posterior component comparison in a vMF mixture, where component-specific concentration prevents directional contrast alone from characterizing the relevant score contrast.

This paper targets posterior-score contrast support through natural parameters formed by multiplying each component direction by its concentration. For each observed coordinate, the component-specific natural parameters are decomposed into a common baseline and a centered contrast. The proposed E-CGL penalty groups the centered contrast across components, so its unit of regularization is the same coordinate-level unit used to define the target support. The common baseline remains unpenalized. An adaptive weighted extension and its path sensitivity are examined separately in the Supplement.

Candidate supports are generated by a guarded path with proximal gradient updates inside generalized EM. They are compared after support-constrained refitting that preserves the target. The primary selector is a practical BIC computed from the refitted likelihood; alternative degrees of freedom and EBIC rules are treated as sensitivity analyses. Support-recovery claims condition on a fixed number of components. When that number is unknown, component-number selection and support selection are examined in separate stages rather than through a single joint criterion.

The empirical study has three parts. Oracle-error calibrated simulations compare posterior-score support recovery across sample size, ambient dimension, and concentration heterogeneity. Shared-canonical and dense-support designs separate the proposed estimand from prototype sparsity and record settings in which sparse posterior-score support is inappropriate. A full-data Classic3 analysis examines clustering, coordinate reduction, and token-level contrasts on a vocabulary-aligned directional representation. Prespecified Classic3 splits assess stability, while BBCSport provides a supplementary scope case.

The contributions are therefore: (i) a posterior-score contrast support estimand for vMF mixtures, its centered representation, and its decomposition into directional and concentration effects; (ii) E-CGL, together with guarded estimation and target-preserving refitting; (iii) existence on the bounded computational parameter space and conditional convergence results for the fixed-responsibility proximal problem and stationary accumulation points under stated regularity conditions; and (iv) an empirical evaluation that distinguishes the intended sparse-support regime from prototype-oriented and dense-support settings. E-CGL is not intended to replace sparse-prototype methods when prototype sparsity is the scientific target.

The remainder of the paper is organized as follows. Section 2 presents the vMF model, posterior-score contrast support, and centered group regularization. Section 3 describes the estimation algorithm, pathwise selection and refitting, and the structural and algorithmic properties used in the analysis. Sections 4 and 5 report the numerical studies and real-data analyses with prespecified preprocessing and split indices, respectively. Section 6 concludes with the empirical scope and limitations of the method.

2. Model and centered regularization

2.1. vMF mixture in natural parameters

Let $x_i \in \mathbb{S}^{d-1}$, $i = 1, \ldots, n$, denote unit-normalized directional observations. A K-component vMF mixture has density

$$f(x_i; \Theta) = \sum_{k=1}^{K} \pi_k C_d(\kappa_k) \exp\{\kappa_k \mu_k^\top x_i\},$$

where $\pi_k > 0$, $\sum_k \pi_k = 1$, $\|\mu_k\|_2 = 1$, and $\kappa_k \geq 0$. Its normalizing constant is

$$C_d(\kappa) = \frac{\kappa^{d/2-1}}{(2\pi)^{d/2} I_{d/2-1}(\kappa)}.$$

At $\kappa = 0$, this expression is interpreted by continuous extension as $C_d(0) = \Gamma(d/2)/(2\pi^{d/2})$.

Write

$$\begin{aligned} &\Theta = (\pi_1, \ldots, \pi_K, \eta_1, \ldots, \eta_K), \\ &E = (\eta_1, \ldots, \eta_K)^\top, \qquad\qquad M = (\mu_1, \ldots, \mu_K)^\top. \end{aligned}$$

where $E, M \in \mathbb{R}^{K \times d}$ and $\eta_k = \kappa_k \mu_k \in \mathbb{R}^d$. Then

$$f(x_i; \Theta) = \sum_{k=1}^{K} \pi_k C_d(\|\eta_k\|_2) \exp(\eta_k^\top x_i).$$

The component-level mapping back to (μ_k, κ_k) and its zero-concentration exception are recorded in Section 3.4. The observed log-likelihood is

$$\ell(\Theta) = \sum_{i=1}^{n} \log \left\{ \sum_{k=1}^{K} \pi_k C_d(\|\eta_k\|_2) \exp(\eta_k^\top x_i) \right\},$$

where the component densities are defined with respect to surface-area measure on $\mathbb{S}^{d-1}$.

2.2. Posterior-score contrast support

The component score

$$s_k(x) = \log \pi_k + \log C_d(\|\eta_k\|_2) + \eta_k^\top x$$

implies the pairwise posterior log-odds representation

$$\log \frac{\Pr(Z = k \mid x)}{\Pr(Z = \ell \mid x)} = a_{k\ell} + (\eta_k - \eta_\ell)^\top x,$$

$$a_{k\ell} = \log \frac{\pi_k}{\pi_\ell} + \log \frac{C_d(\|\eta_k\|_2)}{C_d(\|\eta_\ell\|_2)}.$$

Thus $\eta_k - \eta_\ell$, rather than $\mu_k - \mu_\ell$, is the coefficient of the feature-dependent linear term. For coordinate j, let

$$H_K = I_K - K^{-1}\mathbf{1}_K\mathbf{1}_K^\top, \qquad c_{\cdot j} = H_K \eta_{\cdot j} = \eta_{\cdot j} - \bar{\eta}_j \mathbf{1}_K,$$

where $\bar{\eta}_j = K^{-1}\sum_k \eta_{kj}$. We define

$$S_\eta = \{j : \|c_{\cdot j}\|_2 > 0\} = \bigcup_{k<\ell} \{j : \eta_{kj} - \eta_{\ell j} \neq 0\}.$$

We use *posterior-score contrast support* for this feature-dependent linear support. It does not include the intercepts $a_{k\ell}$. A coordinate with $c_{\cdot j} = 0$ cancels from every pairwise linear term, although its common baseline $\bar{\eta}_j$ remains in the fitted component densities and may affect $a_{k\ell}$ through the concentration norms. Proposition 1 gives the unique centered decomposition and its pairwise-dispersion identity.

For comparison, define

$$S_P = \{j : \|M_{\cdot j}\|_2 > 0\}, \qquad S_\mu = \{j : \|H_K M_{\cdot j}\|_2 > 0\}.$$

The set S_P records coordinates present in at least one directional prototype, whereas S_μ records between-component variation in mean directions. The proposed S_η records coordinates entering posterior-score contrasts. The three targets can differ when component concentrations are heterogeneous.

Accordingly, $c_{\cdot j} \neq 0$ defines a posterior-score contrast coordinate; $c_{\cdot j} = 0$ with $\bar{\eta}_j \neq 0$ defines a common coordinate; and $c_{\cdot j} = 0$ with $\bar{\eta}_j = 0$ defines a globally null coordinate. These categories distinguish the proposed target from prototype support, which records nonzero component coordinates without requiring a difference between components.

The support is invariant to permutations of component labels because such a permutation only reorders the entries of $c_{\cdot j}$. It is not invariant to an arbitrary orthogonal rotation of the feature axes. As with other coordinate-selection procedures, its interpretation is therefore relative to the chosen coordinate system.

2.3. Centered group regularization (E-CGL)

For fixed K, the primary penalty is

$$P_{\mathrm{CGL}}(\eta) = \lambda_\eta \sum_{j=1}^{d} \|H_K \eta_{\cdot j}\|_2 = \lambda_\eta \sum_{j=1}^{d} \|c_{\cdot j}\|_2.$$

Each group is one observed coordinate across the K component contrasts, so the unit of regularization matches the definition of S_η. The common baseline lies in the null space of H_K and is not penalized. The first expression writes the penalty as a function of η, whereas the second displays its centered-contrast target. This coordinate-group construction follows the grouped-variable regularization principle of Yuan and Lin (2006). By contrast,

$$\|\eta_{\cdot j}\|_2^2 = K\bar{\eta}_j^2 + \|c_{\cdot j}\|_2^2,$$

so a raw-η group penalty combines common and contrast effects. A centered direction-space penalty targets S_μ, whereas the Rossi-type entrywise penalty targets S_P. When the κ_k differ, neither target is generally equal to the coefficient support of $\eta_k - \eta_\ell$ in posterior log-odds. These are distinct estimands, not competing parameterizations of the same support.

An entrywise centered penalty selects component-coordinate entries separately: $\lambda_\eta \sum_{k,j} |c_{kj}|$. E-CGL instead selects the complete coordinate contrast as one group. The resulting proximal map and label invariance are stated in Propositions 4 and 3. An adaptive weighted version of this penalty is defined and evaluated in the Supplement. It is treated as a sensitivity extension and is not part of the primary E-CGL estimator or Algorithm 1.

3. Estimation and properties

3.1. Penalized objective and component updates

The finite vMF-mixture likelihood can diverge under concentration collapse (Ng, 2023). We therefore define the fitted model on

$$\mathcal{T}_{\kappa_{\max}} = \left\{(\pi, E) : \pi \in [0,1]^K,\ \mathbf{1}_K^\top \pi = 1,\ \|\eta_k\|_2 \le \kappa_{\max} \text{ for every } k\right\},$$

with $\kappa_{\max} = 10^6$ in all reported analyses. The numerical mixing-weight floor is an implementation safeguard; the existence argument below uses the closed simplex, and the local convergence statement assumes the floor and the concentration bound are inactive.

For fixed K and λ_η, the path-generating criterion is

$$\begin{aligned} \mathcal{L}_{\lambda_\eta}(\Theta) &= \ell(\Theta) - \lambda_\eta \sum_{j=1}^{d} \|H_K \eta_{\cdot j}\|_2 \\ &= \ell(\Theta) - \lambda_\eta \sum_{j=1}^{d} \|c_{\cdot j}\|_2. \end{aligned} \tag{1}$$

The penalty and the vMF normalizing constant are coupled through η_k, so the natural-parameter update is computed by a guarded proximal-gradient generalized M-step.

Lemma 1 (Existence on the finite parameter space)**.** *For fixed K, d, n, finite $\kappa_{\max}$, and nonnegative λ_η, the criterion in* (1) *attains a maximum on $\mathcal{T}_{\kappa_{\max}}$.*

The lemma establishes existence for the bounded computational problem. It does not imply uniqueness, mixture identifiability, or attainment of a global maximum by the numerical algorithm.

3.1.1. E-step and fixed-responsibility subproblem

At outer iteration t, compute $\tau_{ik}^{(t)}$ and

$$\tau_{ik}^{(t)} = \frac{\pi_k^{(t)} C_d(\|\eta_k^{(t)}\|_2) \exp\{(\eta_k^{(t)})^\top x_i\}}{\sum_{h=1}^{K} \pi_h^{(t)} C_d(\|\eta_h^{(t)}\|_2) \exp\{(\eta_h^{(t)})^\top x_i\}}, \tag{2}$$

and define

$$N_k^{(t)} = \sum_{i=1}^{n} \tau_{ik}^{(t)}, \qquad r_k^{(t)} = \sum_{i=1}^{n} \tau_{ik}^{(t)} x_i.$$

Conditional on these responsibilities, the natural-parameter block minimizes

$$F_t(\eta) = f_t(\eta) + g(\eta),$$

$$f_t(\eta) = -\sum_{k=1}^{K} \left\{ (r_k^{(t)})^\top \eta_k + N_k^{(t)} \log C_d(\|\eta_k\|_2) \right\},$$

$$g(\eta) = \lambda_\eta \sum_{j=1}^{d} \|H_K \eta_{\cdot j}\|_2.$$

The smooth term is convex because $-\log C_d(\|\eta_k\|_2)$ is the vMF log-partition function. Its gradient is

$$\nabla_{\eta_k} f_t(\eta) = N_k^{(t)} A_d(\|\eta_k\|_2) \frac{\eta_k}{\|\eta_k\|_2} - r_k^{(t)},$$

with the continuous value $-r_k^{(t)}$ at $\eta_k = 0$, where

$$A_d(\kappa) = \frac{I_{d/2}(\kappa)}{I_{d/2-1}(\kappa)}$$

is the vMF mean-resultant-length function.

3.1.2. Centered proximal-gradient update

Let $J_K = K^{-1}\mathbf{1}_K\mathbf{1}_K^\top$ and $H_K = I_K - J_K$. For a step size $s > 0$, set

$$v = \eta^{(m)} - s\nabla f_t(\eta^{(m)}).$$

The update separates by observed coordinate and has the closed form

$$\eta_{\cdot j}^{(m+1)} = J_K v_{\cdot j} + \left(1 - \frac{s\lambda_\eta}{\|H_K v_{\cdot j}\|_2}\right)_+ H_K v_{\cdot j}. \tag{3}$$

Here $(a)_+ = \max(a, 0)$, and the second term is defined as zero when $H_K v_{\cdot j} = 0$.

The common component is therefore retained, while the centered contrast is group-thresholded; this is a proximal group soft-thresholding operation (Parikh and Boyd, 2014). Mixing proportions are updated by

$$\pi_k^{\text{cand}} = \frac{\max(N_k^{(t)}/n, 10^{-12})}{\sum_h \max(N_h^{(t)}/n, 10^{-12})}.$$

The initial step size is $s = 1/\max_k N_k^{(t)}$. If the smooth majorization condition

$$f_t(\eta^{(m+1)}) \le f_t(\eta^{(m)}) + \langle \nabla f_t(\eta^{(m)}), \eta^{(m+1)} - \eta^{(m)} \rangle + \frac{\|\eta^{(m+1)} - \eta^{(m)}\|_F^2}{2s} \quad (4)$$

is not satisfied within numerical tolerance, s is halved and the proximal proposal is recomputed. This backtracking is repeated for at most 40 trials.

3.2. Guarded proximal generalized-EM algorithm

After the inner proximal-gradient update, the implementation evaluates the penalized auxiliary function and the observed penalized log-likelihood. The candidate is retained only if neither quantity decreases by more than 10^{-8}. Otherwise, the path fit stops, returns the previous accepted estimate, and records a failure diagnostic. Convergence is declared when the relative change in the observed penalized criterion is below 10^{-7}.

Let $Q(\Theta \mid \tau)$ be the expected complete-data log-likelihood and set

$$Q_{\lambda_\eta}(\Theta \mid \tau) = Q(\Theta \mid \tau) - \lambda_\eta \sum_{j=1}^{d} \|H_K \eta_{\cdot j}\|_2.$$

For an old iterate Θ and candidate Θ^+, write

$$\Delta Q_{\lambda_\eta} = Q_{\lambda_\eta}(\Theta^+ \mid \tau) - Q_{\lambda_\eta}(\Theta \mid \tau), \qquad \Delta \mathcal{L}_{\lambda_\eta} = \mathcal{L}_{\lambda_\eta}(\Theta^+) - \mathcal{L}_{\lambda_\eta}(\Theta).$$

These are the two acceptance increments used below.

Components with effective count below 10^{-8} are retained to preserve the fixed value of K and the event is recorded. They are not pruned or reinitialized. These safeguards support statements about accepted numerical updates; they do not establish convergence of the complete parameter sequence, stationarity of every fitted path point, or global optimality.

The recorded analyses use the absolute support threshold $\varepsilon_{\text{supp}} = 10^{-8}$. This threshold is a numerical convention for reading a fitted path support, not an additional penalty parameter.

The recorded simulation and Classic3 controls use at most 200 and 160 proximal-gradient iterations per generalized M-step, respectively, and stop that inner loop when

$$\frac{\|E^{(m+1)} - E^{(m)}\|_F}{\max\{1, \|E^{(m)}\|_F\}} < \varepsilon_{\mathrm{PG}} = 10^{-8}.$$

The smooth majorization tolerance is $\varepsilon_{\mathrm{maj}} = 10^{-10}$, with at most 40 backtracking attempts. The outer generalized-EM loop uses $\varepsilon_{\mathrm{conv}} = 10^{-7}$ and at most 100 iterations, while the auxiliary- and observed-objective acceptance tolerance is $\varepsilon_{\mathrm{acc}} = 10^{-8}$.

Algorithm 1 summarizes the complete E-CGL pathwise procedure.

Algorithm 1. Guarded path algorithm for E-CGL

Input: Data X, component number K, path size $L = 240$, iteration limits, and tolerances $\varepsilon_{\mathrm{PG}}, \varepsilon_{\mathrm{conv}}, \varepsilon_{\mathrm{maj}}, \varepsilon_{\mathrm{acc}}, \varepsilon_{\mathrm{supp}}$.

Output: Selected support $\widehat{S}$, refitted parameters $\widehat{\Theta}^{\mathrm{refit}}_{\widehat{S}}$, and numerical diagnostics.

Stage 1: Dense start and path construction

Dense start: fit the finite-κ dense model from multiple initializations and retain the converged fit with the largest observed log-likelihood.

Path: compute $\widetilde{\lambda}_{\max,\eta}$ from the dense responsibilities and construct the single zero-augmented, KKT-scaled log-geometric path $\Lambda_\eta = \{0, \lambda_1, \ldots, \lambda_{L-1}\}$.

Stage 2: Guarded penalized path

for each $\lambda_\ell \in \Lambda_\eta$ in increasing order **do**; use the preceding valid path fit as a warm start.

  repeat

    E-step: evaluate τ_{ik} by (2); set $N_k \leftarrow \sum_i \tau_{ik}$, $r_k \leftarrow \sum_i \tau_{ik} x_i$.

    Generalized M-step: update π by the floored and renormalized formula in the text; run the inner proximal-gradient iterations until the relative Frobenius-step criterion in the text or the analysis-specific iteration limit is reached.

    Step control: within each inner iteration, halve s and recompute the proximal candidate until (4) holds up to $\varepsilon_{\mathrm{maj}}$, or the 40-attempt backtracking limit is reached.

    Outer guard: after the inner solve, retain Θ^+ if $\Delta Q_{\lambda_\ell} \geq -\varepsilon_{\mathrm{acc}}$ and $\Delta \mathcal{L}_{\lambda_\ell} \geq -\varepsilon_{\mathrm{acc}}$; otherwise mark the path point as failed and terminate the remaining path, without declaring convergence from the unchanged iterate.

  until $|\Delta \mathcal{L}_{\lambda_\ell}|/(1 + |\mathcal{L}_{\lambda_\ell}|) < \varepsilon_{\mathrm{conv}}$ or the 100-iteration limit is reached.

  Store: $S_\ell = \{j : \|H_K \widehat{\eta}_{\cdot j}\|_2 > \varepsilon_{\mathrm{supp}}\}$, the fitted criterion, and diagnostics; stop the path after storing a support of size at most one.

end for

Stage 3: Support-constrained refit and selection

Collect: set $\mathcal{S}_{\mathrm{path}} = \{S_\ell : \lambda_\ell \in \Lambda_\eta\}$ and remove duplicate supports. For a support generated more than once, use the numerically valid path fit with the largest observed log-likelihood as the refit initializer.

Refit: for each $S \in \mathcal{S}_{\mathrm{path}} \cup \{\varnothing\}$, compute $\widehat{\Theta}^{\mathrm{refit}}_S$ subject to $H_K \eta_{\cdot j} = 0$ for $j \notin S$.

Score: $\mathrm{BIC}^{\mathrm{refit}}(S) \leftarrow -2\ell(\widehat{\Theta}^{\mathrm{refit}}_S) + \log(n)\,\mathrm{df}(S)$.

Select: $\widehat{S} \leftarrow \arg\min_S \mathrm{BIC}^{\mathrm{refit}}(S)$.

Return: $\widehat{S}$, $\widehat{\Theta}^{\mathrm{refit}}_{\widehat{S}}$, and diagnostics.

3.3. Pathwise selection and support-constrained refit

3.3.1. Single zero-augmented KKT-scaled path

The path starts at $\lambda_\eta = 0$ and uses warm starts. Responsibilities from the unpenalized dense initializer define (N_k, r_k). Let $r_\bullet = \sum_k r_k$ and $N_\bullet = \sum_k N_k = n$, and let η^{common} be the pooled finite-cap vMF estimate formed from $(N_\bullet, r_\bullet)$. The matrix η_0 repeats η^{common} across components. A KKT-scale upper endpoint for this initial fixed-responsibility subproblem is

$$\widetilde{\lambda}_{\max,\eta} = 1.05 \max_{1\le j\le d} \|H_K \nabla f_t(\eta_0)_{\cdot j}\|_2.$$

This endpoint is a scale for the dense-responsibility convex subproblem, not a global threshold for the observed mixture objective. The single path is

$$\Lambda_\eta = \{0, \lambda_1, \ldots, \lambda_{L-1}\}, \qquad \lambda_\ell = \widetilde{\lambda}_{\max,\eta} 10^{-2+2(\ell-1)/(L-2)}, \quad \ell = 1, \ldots, L-1,$$

with $L = 240$. Thus the positive values range from $0.01\widetilde{\lambda}_{\max,\eta}$ to $\widetilde{\lambda}_{\max,\eta}$ on a logarithmic grid. The path starts at zero and then uses increasing positive penalties with warm starts. The same single path is used for the reported E-CGL simulations and Classic3 analysis. The implementation stops early after storing an active support with at most one coordinate. Independently of this stopping rule, the all-common model is added explicitly as the empty support. The tuning values index candidate supports; the final inferential output is the selected support and its penalty-free constrained refit. The adaptive weighted extension, its augmented-path rerun, and its first-positive support audit are reported only in the Supplement.

3.3.2. Target-preserving support-constrained refit

The penalized path identifies candidate supports, but the final estimates are obtained by a sparsity-penalty-free, finite-cap constrained refit. For a candidate support S, impose

$$j \notin S \quad \Longrightarrow \quad c_{\cdot j} = 0 \quad \Longleftrightarrow \quad \eta_{1j} = \cdots = \eta_{Kj} = \bar{\eta}_j.$$

Thus, an inactive posterior-score coordinate retains an estimated common baseline in the natural-parameter space. The constrained parameterization is

$$\eta_{kj} = b_j + c_{kj}, \qquad \sum_{k=1}^{K} c_{kj} = 0, \qquad c_{kj} = 0 \quad (j \notin S).$$

For each active coordinate, the first $K - 1$ contrasts are free and the last is set to their negative sum. Conditional on the responsibilities, the M-step maximizes

$$Q_S(\eta) = \sum_{k=1}^{K} \left[\eta_k^\top r_k + N_k \log C_d(\|\eta_k\|_2)\right]$$

over $b \in \mathbb{R}^d$ and the free active contrasts. The implementation uses gradient calculations consistent with the numerically evaluated vMF normalizer and L-BFGS-B, followed by step-halving if the observed log-likelihood decreases. Responsibilities, mixing proportions, and the constrained natural parameters are iterated until the numerical stopping criterion is met.

The recorded refit uses at most 160 outer iterations followed, when the first fit is finite and nonfailed but unconverged, by at most 440 additional iterations. Each L-BFGS-B M-step uses at most 80 iterations and projected- gradient tolerance 10^{-6}. The outer refit stops when the absolute log-likelihood increment is no larger than $10^{-7} + 10^{-8}|\ell_{\text{old}}|$.

The target-preserving support-constrained refit enforces the centered support constraint directly while retaining the common baseline. The constrained likelihood is optimized numerically and is not claimed to attain the global maximum of the non-convex mixture likelihood.

The earlier active-only refit imposed $\eta_{\cdot j} = 0$ outside S and therefore removed both the centered contrast and the common baseline. It is retained only as an ablation diagnostic; the target-preserving support-constrained refit is used in the primary analyses.

3.3.3. Support selection by BIC after refit

Let $m_S = |S|$. The practical degrees-of-freedom approximation is

$$\mathrm{df}(S) = d + (K-1)m_S + (K-1)\mathbf{1}(m_S > 0).$$

The three terms count the coordinatewise common baselines, the active centered contrasts, and the mixing proportions when nonzero component contrasts are present. This generic nominal dimension is a practical model-selection approximation, not an exact effective degrees of freedom for a penalized nonregular mixture.

Let

$$\mathcal{S}_{\text{cand}} = \mathcal{S}_{\text{path}} \cup \{\varnothing\}.$$

For each candidate support $S \in \mathcal{S}_{\text{cand}}$, define

$$\text{BIC}^{\text{refit}}(S) = -2\ell(\widehat{\Theta}_S^{\text{refit}}) + \log(n)\,\text{df}(S).$$

The final support is

$$\widehat{S} = \arg\min_{S \in \mathcal{S}_{\text{cand}}} \text{BIC}^{\text{refit}}(S).$$

The all-common model is always compared. Nonempty support candidates remain conditional on the finite path. For a nonempty population support, the event

$$S_\eta^\star \in \mathcal{S}_{\text{path}}$$

defines path coverage. In simulations we separately report path coverage, overall exact recovery, and

$$\Pr\left(\widehat{S} = S_\eta^\star \mid S_\eta^\star \in \mathcal{S}_{\text{path}}\right),$$

the conditional BIC success rate. When $S_\eta^\star = \varnothing$, candidate coverage equals one by construction, and exact recovery is the event $\widehat{S} = \varnothing$. Adding the empty support independently of the recorded path prevents the stopping rule from omitting the all-common model. An audit of the previously recorded paths showed that this addition left all primary and estimand-separation conclusions unchanged, but selected the empty support in a subset of stress-condition repetitions; those stress summaries were recomputed under the corrected candidate family.

When several numerically valid path fits generate the same support, the fit with the largest observed log-likelihood initializes the single constrained refit. This rule is applied before support-level BIC comparison and does not use the external class labels.

The tuning parameter is therefore treated only as a candidate-generation index. BIC after refit selects a distinct support and does not define or require a uniquely interpreted estimate of λ_η.

All distinct nonempty supports encountered on the path and the explicitly added empty support are refitted. The information criterion follows the BIC principle of Schwarz (1978), using the generic nominal dimension above. It is a within-method support selector, not an exact marginal-likelihood approximation or a consistency result for a singular mixture family.

Each competing procedure has its own support target and nominal dimension. BIC therefore selects refitted candidates separately within each method.

Values across methods are descriptive and do not provide a common-scale ranking of the support targets.

An EBIC sensitivity criterion is

$$\mathrm{EBIC}_\gamma(S) = \mathrm{BIC}^{\mathrm{refit}}(S) + 2\gamma \log \binom{d}{m_S}, \qquad \gamma \in \{0.25, 0.5, 1\}.$$

EBIC (Chen and Chen, 2008) and alternative degrees-of-freedom rules are treated as sensitivity analyses rather than changes to the primary selector.

3.3.4. Selection of the number of components

Component-number selection is not part of the E-CGL support estimator. The number of components and the sparsity level are therefore selected in separate stages. In support-recovery simulations, K is fixed at the data-generating value so that support selection is not confounded with component-number misspecification.

For an unknown K, let $\mathcal{K}$ be a prespecified candidate set. The practical two-stage procedure is:

1. Fit an unpenalized dense vMF mixture for each $K \in \mathcal{K}$ using multiple starts.
2. Compare candidate K values using label-free density criteria, information criteria, and partition stability as appropriate to the application.
3. Fix the selected component resolution $\widehat{K}$.
4. Fit the E-CGL path at $\widehat{K}$ and choose a distinct support by BIC after the target-preserving support-constrained refit.

Symbolically,

$$\widehat{K} = \arg\min_{K \in \mathcal{K}} \mathcal{C}_{\mathrm{dense}}(K), \qquad \widehat{S}_{\widehat{K}} = \arg \min_{S \in \mathcal{S}_{\mathrm{cand}}(\widehat{K})} \mathrm{BIC}^{\mathrm{refit}}(\widehat{K}, S).$$

Here $\mathcal{C}_{\mathrm{dense}}$ may use independent or held-out negative log-likelihood, bootstrap out-of-bag negative log-likelihood, or an information criterion. No single information criterion is asserted to select the correct resolution in every high-dimensional mixture. When density fit and stability favor different K, both competing resolutions are reported.

3.3.5. Computational implementation

The implementation uses log-sum-exp stabilization in the E-step, warm starts along the penalty path, and explicit convergence and constraint checks in the centered-support refit. Optional Rcpp helpers replace low-level numerical operations without changing the statistical algorithm; R-only and Rcpp-helper outputs were checked to numerical tolerance before runtime comparisons.

The current computational claims are limited to empirical equality and runtime diagnostics. The method does not rely on an Rcpp implementation for its statistical definition.

The finite parameter space $\mathcal{T}_{\kappa_{\max}}$ is used in dense initialization, the penalized path, and support-constrained refitting. Backtracking rejects any proposed E-CGL iterate outside this space. A converged refit that reaches 99% of the bound is retained for diagnosis but is not eligible for information-criterion selection. This guard defines the finite computational problem; it is not an additional sparsity penalty. Target-preserving refitting refers to enforcement of the centered-support constraint, not to a global mixture optimum.

3.4. Structural properties

The following statements justify the centered support estimand for a fixed number of components. They do not imply uniqueness of a fitted mixture, which remains subject to finite-mixture identifiability and label permutation. Full proofs are given in Appendix A.

Proposition 1 (Centered decomposition and pairwise support)**.** *For any $v \in \mathbb{R}^K$, there is a unique decomposition*

$$v = \bar{v}\mathbf{1}_K + c, \qquad \bar{v} = K^{-1}\mathbf{1}_K^\top v, \qquad \mathbf{1}_K^\top c = 0.$$

For $v = \eta_{\cdot j}$, the following statements are equivalent:

$$c = 0; \qquad \eta_{1j} = \cdots = \eta_{Kj}; \qquad \eta_{kj} - \eta_{\ell j} = 0 \quad \text{for all } k < \ell.$$

Moreover,

$$\|c\|_2^2 = \frac{1}{K}\sum_{k<\ell}(\eta_{kj} - \eta_{\ell j})^2.$$

Consequently,

$$S_\eta = \bigcup_{k<\ell} \{j : \eta_{kj} - \eta_{\ell j} \neq 0\},$$

the union of the feature-dependent linear supports in all pairwise posterior log-odds.

Proposition 1 clarifies the scope of the estimand. If $c_{\cdot j} = 0$, the coefficient of x_j cancels from every pairwise posterior log-odds. The common baseline is nevertheless retained in η_k and may affect the intercept through $\|\eta_k\|_2$; it is therefore not irrelevant to the component density.

Proposition 2 (Canonical–directional relation)**.** *Let $\kappa = (\kappa_1, \ldots, \kappa_K)^\top$, $\bar\kappa = K^{-1}\mathbf{1}_K^\top\kappa$, and $h = \kappa - \bar\kappa\mathbf{1}_K$. Write*

$$\bar\mu = K^{-1}M^\top\mathbf{1}_K, \qquad U = H_K M.$$

Then $M = \mathbf{1}_K\bar\mu^\top + U$ and

$$H_K E = \bar\kappa U + h\bar\mu^\top + H_K \operatorname{diag}(h)U.$$

The three terms represent directional heterogeneity, concentration heterogeneity along the common direction, and their interaction.

If $\kappa_1 = \cdots = \kappa_K = \kappa > 0$, then $H_K E = \kappa H_K M$ and $S_\eta = S_\mu$. If $\mu_1 = \cdots = \mu_K = \bar\mu$ and $h \neq 0$, then $S_\mu = \varnothing$ while $S_\eta = \operatorname{supp}(\bar\mu)$. Conversely, heterogeneous concentrations can make $H_K M_{\cdot j} \neq 0$ even when $H_K E_{\cdot j} = 0$. In the general decomposition the three terms may cancel, so S_η is not identified with the union of their separate supports.

Proposition 3 (Label equivariance and support invariance)**.** *Let P be a $K \times K$ permutation matrix. Then*

$$H_K P = P H_K, \qquad \|H_K P \eta_{\cdot j}\|_2 = \|H_K \eta_{\cdot j}\|_2.$$

The E-CGL penalty is invariant to component relabeling, its proximal map is equivariant, and S_η is unchanged.

For a nonzero natural parameter, the component representation is recovered as $\kappa_k = \|\eta_k\|_2$ and $\mu_k = \eta_k/\|\eta_k\|_2$. At $\eta_k = 0$, the vMF component is uniform and its direction is not identified. These algebraic statements do not establish finite-mixture identifiability or uniqueness of the fitted mixture.

Component-specific interpretation assumes distinct component distributions and positive mixing weights and is made only up to label permutation. If two components have identical natural parameters, their individual mixing weights are not separately identified. Support-recovery statements are conditional on fixed K.

3.5. Algorithmic properties

Proposition 4 (Closed-form proximal update). *Let $J_K = K^{-1}\mathbf{1}_K\mathbf{1}_K^\top$ and $H_K = I_K - J_K$. For $a \geq 0$ and $v \in \mathbb{R}^K$,*

$$\operatorname{prox}_{a\|H_K\cdot\|_2}(v) = J_K v + \left(1 - \frac{a}{\|H_K v\|_2}\right)_+ H_K v,$$

where the second term is defined as zero when $H_K v = 0$.

The common baseline is unchanged. Group soft thresholding is applied only to the centered contrast, matching the coordinate-level definition of S_η.

Proposition 5 (Smooth majorization and proximal descent). *Fix responsibilities with $N_k > 0$. The function f_t in Section 3.1.1 is convex and has a Lipschitz gradient with constant $L \leq \max_k N_k$. Hence, for $s \leq 1/L$, the proximal-gradient update in Section 3.1.2 does not increase F_t in exact arithmetic. More generally, an update accepted with majorization tolerance $\varepsilon_{\mathrm{maj}}$ satisfies*

$$F_t(\eta^{(m+1)}) \leq F_t(\eta^{(m)}) + \varepsilon_{\mathrm{maj}}.$$

Proposition 5 motivates the backtracking rule. The implementation starts from $s = 1/\max_k N_k$ and halves the step until the majorization inequality is satisfied numerically. This check also protects against error from numerical evaluation of the vMF normalizing constant.

Theorem 1 (Fixed-responsibility proximal convergence). *Let $d \geq 2$. Fix responsibilities such that $N_k > 0$ and $\|r_k\|_2 < N_k$ for every k, fix a finite $\lambda_\eta \geq 0$. Then F_t on $\mathbb{R}^{K\times d}$ is strictly convex and coercive and therefore has a unique unconstrained minimizer $E_t^\star$. If $E_t^\star$ lies strictly inside the concentration bound, it is also the unique solution of the bounded subproblem. Exact unconstrained proximal gradient with the stated majorization backtracking rule converges to $E_t^\star$, and its objective gap is $O(m^{-1})$ after m inner iterations.*

The theorem concerns one fixed-responsibility subproblem. The reported code terminates the inner loop at a fixed relative-step tolerance or iteration limit, so a finite fitted path is an inexact numerical approximation to this limiting sequence. It describes the bounded implementation only when the cap is inactive along the inner sequence. An active cap would require a separate projected-proximal analysis (Beck and Teboulle, 2009; Parikh and Boyd, 2014).

For the current responsibilities, recall

$$Q_{\lambda_\eta}(\Theta \mid \tau^{(t)}) = Q(\Theta \mid \tau^{(t)}) - \lambda_\eta \sum_{j=1}^{d} \|H_K \eta_{\cdot j}\|_2.$$

Proposition 6 (Penalized generalized-M monotonicity)**.** *Let $\tau^{(t)}$ be the posterior responsibilities under $\Theta^{(t)}$. If a candidate $\Theta^{(t+1)}$ satisfies*

$$Q_{\lambda_\eta}(\Theta^{(t+1)} \mid \tau^{(t)}) \geq Q_{\lambda_\eta}(\Theta^{(t)} \mid \tau^{(t)}),$$

then, in exact arithmetic,

$$\mathcal{L}_{\lambda_\eta}(\Theta^{(t+1)}) \geq \mathcal{L}_{\lambda_\eta}(\Theta^{(t)}).$$

The implementation verifies both inequalities directly and permits decreases of at most 10^{-8}, so its finite trace is only numerically near-monotone. For the exact-arithmetic algorithm with zero acceptance tolerance, a bounded penalized criterion has a nondecreasing and therefore convergent sequence of objective values. This monotonicity result alone does not establish boundedness or convergence of the full parameter sequence, stationary-point convergence, or global optimality. Theorem 2 gives a separate conditional stationarity result under explicit residual assumptions.

For a responsibility array τ, let f_τ denote the smooth fixed-responsibility objective in Section 3.1.1, and write

$$P(E) = \sum_{j=1}^{d} \|H_K E_{\cdot j}\|_2.$$

For $s > 0$, define the natural-parameter proximal-gradient mapping

$$\mathcal{G}_s(E; \tau) = \frac{1}{s}\left[E - \operatorname{prox}_{s\lambda_\eta P}\{E - s\nabla f_\tau(E)\}\right],$$

and define the exact interior mixing-weight residual by

$$\mathcal{R}_\pi(\pi;\tau) = \pi - \frac{N(\tau)}{n}, \qquad N_k(\tau) = \sum_{i=1}^{n} \tau_{ik}.$$

For Theorem 2, s_t is the diagnostic step size used to evaluate the natural-parameter residual at the final accepted inner iterate; it need not equal every trial step used during backtracking.

Theorem 2 (Stationarity under vanishing block residuals)**.** *Fix λ_η. Consider an idealized generalized-EM sequence $\{\Theta^{(t)}\}$ satisfying the following conditions: (i) the E-step is exact, so $\tau^{(t)} = \tau(\Theta^{(t)})$; (ii) the iterates and their accepted successors lie in a compact subset of the interior of the simplex and concentration bound, with $\inf_{t,k} N_k(\tau^{(t)}) > 0$; (iii) the inner step sizes satisfy $0 < \underline{s} \le s_t \le \overline{s} < \infty$; (iv) the sequence is asymptotically regular, $\|\Theta^{(t+1)} - \Theta^{(t)}\| \to 0$; and (v)*

$$\|\mathcal{G}_{s_t}(E^{(t+1)};\tau^{(t)})\|_F + \|\mathcal{R}_\pi(\pi^{(t+1)};\tau^{(t)})\|_2 \longrightarrow 0.$$

Then every accumulation point $\bar{\Theta} = (\bar{\pi}, \bar{E})$ satisfies the interior KKT conditions

$$0 \in -\nabla_E \ell(\bar{\Theta}) + \lambda_\eta \partial P(\bar{E}),$$

and, for some scalar ν,

$$\nabla_\pi \ell(\bar{\Theta}) = \nu \mathbf{1}_K, \qquad \mathbf{1}_K^\top \bar{\pi} = 1.$$

Thus every accumulation point is stationary for the finite-parameter penalized observed objective. The stationarity statement is equivariant under component-label permutation.

This theorem is a conditional local result for the explicitly stated idealized sequence, not a convergence claim for every finite-tolerance path. Fixed stopping tolerances, failed path points, active numerical bounds, absence of asymptotic regularity, or component degeneracy fall outside its assumptions; the empirical diagnostics for those cases are reported separately (Dempster et al., 1977; Green, 1990; Wu, 1983; Tseng, 2004).

3.6. Computational complexity

For dense X, one E-step costs $O(nKd)$; for a sparse matrix it costs $O\{K\,\mathrm{nnz}(X)\}$. Forming the sufficient statistics has the same order, and one centered proximal-gradient update costs $O(Kd)$. Excluding the input matrix, the main working memory is $O(nK+Kd)$, in addition to $O(nd)$ for dense X or $O\{\mathrm{nnz}(X)\}$ for sparse X. If the single path evaluates L tuning points, T_{EM} outer iterations, and T_{PG} inner updates on average, its leading cost is

$$O\left[LT_{\mathrm{EM}}\left\{K\,\mathrm{nnz}(X)+T_{\mathrm{PG}}Kd\right\}\right]$$

for sparse input, plus the costs of distinct-support refits. These are operation counts rather than runtime superiority claims.

4. Numerical studies

4.1. Design and evaluation criteria

The primary experiment evaluates recovery of posterior-score contrast support under varying overlap and concentration heterogeneity. Observations are drawn from a balanced $K=4$ vMF mixture in $d=200$ dimensions. The coordinates are partitioned as

$$\{1,\ldots,d\}=S_C\cup S_D\cup S_N,\qquad (q_C,q_D,q_N)=(4,16,180),$$

where S_C contains common natural-parameter coefficients, S_D contains centered component contrasts, and S_N is zero. The clustering difficulty is calibrated by the oracle Bayes error

$$e_B=Pr_{\Theta^\star}\left\{\arg\max_k \Pr(Z=k\mid X;\Theta^\star)\neq Z\right\}.$$

We use $e_B\in\{0.025,0.05,0.10\}$, $n\in\{300,1000\}$, and either

$$\kappa=(45,45,45,45)\quad\text{or}\quad\kappa=(30,40,50,60).$$

The common-to-contrast allocation is calibrated by Monte Carlo bisection while the prescribed concentrations and support sets are held fixed. Independent validation samples of size 200,000 gave achieved errors between 0.02566 and 0.10035; the largest absolute discrepancy from its target was 0.00172.

Table 1: Methods and roles in the numerical studies.

Method	support target	role
Spherical k-means	none	external clustering baseline
Dense vMF, shared κ	none	concentration-restricted baseline
Dense vMF, free κ_k	none	likelihood baseline
M-L	S_P	published prototype-sparsity reference
M-CGL	S_μ	directional companion
E-CGL	S_η	primary proposed estimator

Each of the 12 cells contains 100 paired repetitions and an independent test sample of size 5,000. The common comparison set consists of spherical k-means, dense vMF mixtures with shared and component-specific concentrations, the Rossi-type entrywise prototype lasso (M-L), and E-CGL. M-CGL is included as a directional companion in the four prespecified $e_B = 0.05$ cells. M-L and E-CGL use single 240-point paths, while M-CGL uses the union of supports from 60- and 120-point paths. All mixture fits use 10 dense starts, Rcpp helpers, and BIC after target-specific support refitting. The E-ACGL adaptive sensitivity results, including their augmented rerun and path-coverage audit, are reported in the Supplement and are not part of the primary comparison set.

Table 1 records the estimand and role of each method. The common comparison set is evaluated in all 12 primary cells; M-CGL is restricted to the four prespecified $e_B = 0.05$ companion cells.

For clarity, the two mean-direction comparators use different penalties. The Rossi-type entrywise prototype penalty is

$$P_{\mathrm{M-L}}(M) = \lambda_\mu \sum_{k=1}^{K} \sum_{j=1}^{d} |\mu_{kj}|, \qquad \|\mu_k\|_2 = 1,$$

whereas the directional companion uses

$$P_{\mathrm{M-CGL}}(M) = \lambda_\mu \sum_{j=1}^{d} w_j^{(\mu)} \|H_K M_{\cdot j}\|_2, \qquad \|\mu_k\|_2 = 1.$$

The primary simulations estimate component-specific concentrations for M-L. The Rossi bridge and the real-data M-L analysis use a shared concentration, as prespecified for those comparisons. Full M-CGL optimization details are given in the Supplement.

Clustering is evaluated by ARI and NMI. Predictive fit is evaluated by the per-observation test negative log-likelihood

$$\mathrm{NLL}_{\mathrm{test}} = -\frac{1}{n_{\mathrm{test}}}\sum_{i=1}^{n_{\mathrm{test}}} \log f(X_i; \widehat{\Theta}),$$

where f is the vMF-mixture density with respect to surface-area measure on the unit sphere. Such a density can exceed one, so its reported NLL can be negative. Posterior-score support recovery is summarized by

$$\mathrm{TPR}_\eta, \quad \mathrm{FPR}_\eta, \quad F_{1,\eta}, \quad \Pr(\widehat{S}_\eta = S_\eta),$$

and centered natural-parameter estimation by

$$\mathrm{MSE}_{\eta^c} = \frac{1}{Kd}\sum_{k=1}^{K}\sum_{j=1}^{d}(\widehat{c}_{kj} - c^\star_{kj})^2,$$

after label alignment. Each sparse procedure is also evaluated against its own population support. Consequently, $F_{1,P}$, $F_{1,\mu}$, and $F_{1,\eta}$ are target-specific quantities rather than a common ranking.

4.2. Posterior-score support recovery

Tables 2 and 3 report every prespecified combination of oracle Bayes error, concentration pattern, and method, with estimation, posterior-score selection, and clustering quantities shown in separate column blocks. Figure 1 shows the repetition-level ARI distributions, and Figure 2 shows the corresponding $\log_{10}(\mathrm{MSE}_{\eta^c})$ distributions for the four likelihood-based methods over the complete 12-cell grid. In these common-target comparisons, support metrics are evaluated against S_η; M-CGL is evaluated against its own directional target in Section 4.3.

Across 11 of the 12 primary cells, E-CGL selected between 16.02 and 16.60 coordinates and attained $F_{1,\eta} \geq 0.983$. The exception combined $n = 300$, $e_B = 0.10$, and heterogeneous concentrations. In that cell E-CGL selected 24.34 coordinates and attained $F_{1,\eta} = 0.768$. At $n = 1000$, E-CGL had $F_{1,\eta} \geq 0.986$ in all six concentration–overlap cells. M-L retained almost all 200 coordinates and consequently had $F_{1,\eta}$ near 0.148. ARI decreased as e_B increased even when support recovery remained accurate, reflecting the calibrated increase in population overlap. Supplementary Tables S1–S2 add support decomposition, NMI, Monte Carlo standard errors for all reported means, failure summaries, and the E-ACGL sensitivity results.

Table 2: Primary method comparison at $n = 300$. Estimation, posterior-score selection, and clustering performance are reported over 100 paired repetitions.

			Estimation	Posterior-score selection				Clustering and fit	
e_B	κ pattern	method	MSE_{η^c} (100 × MCSE)	$\widehat{q}$	FPR_η	TPR_η	$F_{1,\eta}$	ARI	test NLL
0.025	equal	Sph. k-means	–	–	–	–	–	0.863	–
		Dense-S	2.471 (1.652)	–	–	–	–	0.869	-245.921
		Dense-F	2.505 (1.675)	–	–	–	–	0.864	-245.906
		M-L	2.447 (1.675)	199.58	0.998	1.000	0.148	0.865	-245.934
		E-CGL	0.198 (0.402)	16.14	0.001	1.000	0.996	0.927	-246.935
0.025	heterogeneous	Sph. k-means	–	–	–	–	–	0.775	–
		Dense-S	3.092 (1.986)	–	–	–	–	0.798	-245.869
		Dense-F	2.540 (1.702)	–	–	–	–	0.856	-246.110
		M-L	2.475 (1.648)	199.61	0.998	1.000	0.148	0.860	-246.138
		E-CGL	0.192 (0.512)	16.15	0.001	1.000	0.996	0.925	-247.153
0.050	equal	Sph. k-means	–	–	–	–	–	0.620	–
		Dense-S	2.810 (2.275)	–	–	–	–	0.738	-245.936
		Dense-F	2.863 (2.347)	–	–	–	–	0.726	-245.919
		M-L	2.771 (2.133)	199.65	0.998	1.000	0.148	0.731	-245.954
		M-CGL	0.222 (0.588)	16.23	0.001	1.000	0.993	0.856	-247.010
		E-CGL	0.221 (0.599)	16.26	0.001	1.000	0.992	0.856	-247.010
0.050	heterogeneous	Sph. k-means	–	–	–	–	–	0.580	–
		Dense-S	3.764 (4.549)	–	–	–	–	0.649	-245.860
		Dense-F	3.077 (4.439)	–	–	–	–	0.709	-246.082
		M-L	2.957 (4.432)	199.70	0.998	1.000	0.148	0.718	-246.122
		M-CGL	0.628 (2.619)	20.73	0.033	0.912	0.797	0.812	-247.060
		E-CGL	0.229 (0.778)	16.60	0.003	1.000	0.983	0.857	-247.206
0.100	equal	Sph. k-means	–	–	–	–	–	0.211	–
		Dense-S	4.191 (9.094)	–	–	–	–	0.436	-245.788
		Dense-F	4.509 (10.058)	–	–	–	–	0.390	-245.737
		M-L	4.143 (9.850)	199.67	0.998	1.000	0.148	0.422	-245.830
		E-CGL	0.253 (0.673)	16.28	0.002	1.000	0.992	0.724	-247.130
0.100	heterogeneous	Sph. k-means	–	–	–	–	–	0.444	–
		Dense-S	4.768 (4.502)	–	–	–	–	0.484	-245.998
		Dense-F	4.591 (7.985)	–	–	–	–	0.512	-246.148
		M-L	4.529 (8.296)	199.62	0.998	1.000	0.148	0.514	-246.173
		E-CGL	0.892 (2.925)	24.34	0.050	0.943	0.768	0.638	-247.138

Notes: Point estimates are reported on their original scales; parenthesized values are Monte Carlo standard errors of mean MSE_{η^c} multiplied by 100 and reported to three decimal places. They are not standard errors of fitted parameters. Selection rates are on the 0–1 scale and use S_η for all sparse methods. Target-specific M-CGL results are reported separately. Dashes indicate that a method does not return the corresponding quantity. M-CGL was prespecified only for $e_B = 0.05$.

Table 3: Primary method comparison at $n = 1000$. Estimation, posterior-score selection, and clustering performance are reported over 100 paired repetitions.

			Estimation	Posterior-score selection				Clustering and fit	
e_B	κ pattern	method	MSE_{η^c} (100 × MCSE)	$\widehat{q}$	FPR_η	TPR_η	$F_{1,\eta}$	ARI	test NLL
0.025	equal	Sph. k-means	–	–	–	–	–	0.917	–
		Dense-S	0.676 (0.347)	–	–	–	–	0.918	-246.966
		Dense-F	0.679 (0.342)	–	–	–	–	0.917	-246.964
		M-L	0.672 (0.339)	199.42	0.997	1.000	0.149	0.918	-246.968
		E-CGL	0.055 (0.110)	16.02	0.000	1.000	0.999	0.933	-247.245
0.025	heterogeneous	Sph. k-means	–	–	–	–	–	0.866	–
		Dense-S	1.147 (0.534)	–	–	–	–	0.861	-246.931
		Dense-F	0.685 (0.404)	–	–	–	–	0.912	-247.167
		M-L	0.677 (0.405)	199.57	0.998	1.000	0.148	0.913	-247.170
		E-CGL	0.056 (0.115)	16.02	0.000	1.000	0.999	0.928	-247.451
0.050	equal	Sph. k-means	–	–	–	–	–	0.835	–
		Dense-S	0.731 (0.436)	–	–	–	–	0.837	-247.035
		Dense-F	0.737 (0.437)	–	–	–	–	0.836	-247.032
		M-L	0.726 (0.437)	199.58	0.998	1.000	0.148	0.837	-247.036
		M-CGL	0.061 (0.150)	16.04	0.000	1.000	0.999	0.867	-247.316
		E-CGL	0.061 (0.161)	16.06	0.000	1.000	0.998	0.867	-247.316
0.050	heterogeneous	Sph. k-means	–	–	–	–	–	0.773	–
		Dense-S	1.273 (0.642)	–	–	–	–	0.765	-247.004
		Dense-F	0.747 (0.476)	–	–	–	–	0.835	-247.228
		M-L	0.736 (0.478)	199.67	0.998	1.000	0.148	0.836	-247.232
		M-CGL	0.087 (0.143)	20.04	0.022	1.000	0.888	0.867	-247.510
		E-CGL	0.061 (0.132)	16.02	0.000	1.000	0.999	0.869	-247.517
0.100	equal	Sph. k-means	–	–	–	–	–	0.676	–
		Dense-S	0.858 (0.482)	–	–	–	–	0.684	-247.146
		Dense-F	0.866 (0.490)	–	–	–	–	0.682	-247.144
		M-L	0.851 (0.492)	199.64	0.998	1.000	0.148	0.683	-247.149
		E-CGL	0.067 (0.133)	16.02	0.000	1.000	0.999	0.747	-247.445
0.100	heterogeneous	Sph. k-means	–	–	–	–	–	0.589	–
		Dense-S	1.758 (1.744)	–	–	–	–	0.595	-247.110
		Dense-F	1.013 (1.822)	–	–	–	–	0.669	-247.308
		M-L	0.970 (1.583)	199.88	0.999	1.000	0.148	0.675	-247.316
		E-CGL	0.078 (0.227)	16.47	0.003	1.000	0.986	0.753	-247.627

Notes: Point estimates are reported on their original scales; parenthesized values are Monte Carlo standard errors of mean MSE_{η^c} multiplied by 100 and reported to three decimal places. They are not standard errors of fitted parameters. Selection rates are on the 0–1 scale and use S_η for all sparse methods. Target-specific M-CGL results are reported separately. Dashes indicate that a method does not return the corresponding quantity. M-CGL was prespecified only for $e_B = 0.05$.

Table 4: Estimand-separation diagnostics over 100 repetitions. Support F1 is reported against both population targets.

design	method	q^*_μ	q^*_η	$\widehat{q}$	$F_{1,\mu}$	$F_{1,\eta}$	exact own target	ARI
common κ	M-CGL	16	16	16.04	0.999	0.999	0.96	0.867
common κ	E-CGL	16	16	16.06	0.998	0.998	0.95	0.867
pure concentration	M-CGL	0	16	0.77	–	0.004	0.24	0.674
pure concentration	E-CGL	0	16	16.33	0.000	0.990	0.77	0.635
shared canonical	M-CGL	100	20	21.16	0.348	0.973	0.00	0.859
shared canonical	E-CGL	100	20	20.02	0.333	1.000	0.98	0.870
crossed support	M-CGL	12	12	11.67	0.983	0.677	0.64	0.996
crossed support	E-CGL	12	12	11.87	0.681	0.980	0.69	0.996

4.3. Directional versus posterior-score estimands

Four diagnostics separate directional and posterior-score heterogeneity. Under common concentration, $S_\mu = S_\eta$. Under pure concentration heterogeneity, the component directions are common, so $S_\mu = \varnothing$ while S_η contains 16 coordinates. The shared-canonical design contains 80 common natural-parameter coordinates and 20 posterior-score coordinates. The crossed design contains four η-only, four μ-only, eight jointly active, and eight null coordinates.

For the empty directional target, "exact own target" is the probability of selecting the empty support. M-CGL selected that support in 24% of the pure- concentration repetitions, whereas E-CGL recovered the nonempty posterior-score support with mean $F_{1,\eta} = 0.990$. In the crossed design, each method had high F1 against its own target and lower F1 against the other target. The shared-canonical design exposed a boundary for the directional companion: its exact support did not occur on the recorded path. These results motivate target-specific, rather than winner-takes-all, comparisons.

4.4. Sample-size recovery and oracle benchmarks

Table 5 gives the complete numerical trajectories and Figure 3 displays their trends. E-CGL is followed over $n \in \{300, 600, 1000, 2000\}$. Both $e_B = 0.05$ and $e_B = 0.10$ include equal and heterogeneous concentration trajectories. At $e_B = 0.05$, centered-η MSE decreased from 0.221 to 0.028 under equal concentrations and from 0.229 to 0.028 under heterogeneous concentrations. FPR decreased to zero and exact-support recovery reached one at $n = 2000$. Under $e_B = 0.10$ and equal concentrations, E-CGL moved from $\widehat{q} = 16.28$, $F_{1,\eta} = 0.992$, and exact recovery 0.76 at $n = 300$ to $\widehat{q} = 16.00$,

$F_{1,\eta} = 1.000$, and exact recovery 1.00 at $n = 2000$; MSE_{η^c} decreased from 0.253 to 0.031. Under the heterogeneous counterpart, E-CGL moved from $\widehat{q} = 24.34$, $F_{1,\eta} = 0.768$, and exact recovery 0.01 at $n = 300$ to $\widehat{q} = 16.03$, $F_{1,\eta} = 0.999$, and exact recovery 0.97 at $n = 2000$.

The path oracle is the fitted candidate with the largest target-specific F1 within each repetition. The oracle-S_η estimator refits under the known population support. We report

$$\Delta_{\mathrm{sel}} = F_1^{\text{path oracle}} - F_1^{\mathrm{BIC}}, \qquad \Delta_{\mathrm{NLL}} = \mathrm{NLL}(\widehat{\Theta}) - \mathrm{NLL}(\widehat{\Theta}_{S_\eta}^{\text{oracle support}}).$$

These finite-sample benchmark gaps are not claims about asymptotic support recovery. At $e_B = 0.05$, all E-CGL selector gaps were at most 0.0079 at $n = 300$ and at most 0.0018 for $n \geq 600$. The difficult heterogeneous-κ, $e_B = 0.10$ trajectory had $\Delta_{\mathrm{sel}} = 0.046$ and $\Delta_{\mathrm{NLL}} = 0.191$ at $n = 300$. The NLL gap decreased with sample size; the selector gap rose slightly at $n = 600$ before decreasing to 0.0009 at $n = 2000$. The BIC, path-oracle, and oracle-support-refit comparisons are in Supplementary Table S4.

4.5. Stress conditions and computation

Five stress cells examine moderately dense support at $n = 300$ and $n = 1000$, strongly dense support, $d > n$, and a weak beta-min condition. In the sparse weak-signal cell, E-CGL retained mean $F_{1,\eta} = 0.998$ and exact recovery 0.93. In the $d = 500, n = 300$ cell, E-CGL's F1 was 0.796 and exact recovery 0.01. For 80 active coordinates at $n = 300$, the explicit empty support had the smallest BIC in 81 of 100 E-CGL repetitions. Consequently, E-CGL selected 15.25 coordinates on average, with $F_{1,\eta} = 0.154$ and ARI 0.105. At $n = 1000$, E-CGL did not select the empty support and attained $F_{1,\eta} = 0.890$. This contrast identifies repeated selection of the all-common model by the practical refit BIC, rather than path failure alone, as the principal small-sample limitation in the moderately dense cell. With 160 active coordinates, E-CGL selected 153.96 on average and had $F_{1,\eta} = 0.924$. Full E-CGL stress results and method-phase convergence summaries are reported in Table 6; Supplementary Tables S5–S6 add test NLL, Monte Carlo standard errors, convergence, path-endpoint, timing diagnostics, and the separate E-ACGL sensitivity results.

In a paired runtime diagnostic for the representative heterogeneous $e_B = 0.05$, $n = 1000$, $d = 200$ cell, median end-to-end wall-clock times (IQR) were 383.02 (343.71–418.68) seconds for M-CGL, 48.31 (46.36–49.85) seconds for

Table 5: E-CGL sample-size recovery over 100 repetitions. Parenthesized values are Monte Carlo standard errors multiplied by 100.

Panel A: centered natural-parameter contrast estimation

e_B	κ pattern	n	$\widehat{q}$	MSE_{η^c} (100 × MCSE)	$n\,\mathrm{MSE}_{\eta^c}$
0.05	equal	300	16.26	0.221 (0.599)	66.37
		600	16.02	0.097 (0.215)	58.24
		1000	16.06	0.061 (0.161)	61.26
		2000	16.00	0.028 (0.055)	56.45
0.05	heterogeneous	300	16.60	0.229 (0.778)	68.69
		600	16.06	0.099 (0.245)	59.68
		1000	16.02	0.061 (0.132)	61.40
		2000	16.00	0.028 (0.061)	55.64
0.10	equal	300	16.28	0.253 (0.673)	75.96
		600	16.03	0.110 (0.243)	66.19
		1000	16.02	0.067 (0.133)	66.56
		2000	16.00	0.031 (0.063)	62.76
0.10	heterogeneous	300	24.34	0.892 (2.925)	267.71
		600	18.37	0.198 (1.025)	118.60
		1000	16.47	0.078 (0.227)	78.22
		2000	16.03	0.035 (0.079)	69.95

Panel B: posterior-score support and clustering

e_B	κ pattern	n	TPR_η (100 × MCSE)	FPR_η (100 × MCSE)	$F_{1,\eta}$	exact support	ARI
0.05	equal	300	1.000 (0.000)	0.001 (0.029)	0.992	0.77	0.856
		600	1.000 (0.000)	0.000 (0.008)	0.999	0.98	0.864
		1000	1.000 (0.000)	0.000 (0.015)	0.998	0.95	0.867
		2000	1.000 (0.000)	0.000 (0.000)	1.000	1.00	0.869
0.05	heterogeneous	300	1.000 (0.000)	0.003 (0.064)	0.983	0.69	0.857
		600	1.000 (0.000)	0.000 (0.015)	0.998	0.95	0.866
		1000	1.000 (0.000)	0.000 (0.008)	0.999	0.98	0.869
		2000	1.000 (0.000)	0.000 (0.000)	1.000	1.00	0.871
0.10	equal	300	1.000 (0.000)	0.002 (0.029)	0.992	0.76	0.724
		600	1.000 (0.000)	0.000 (0.009)	0.999	0.97	0.742
		1000	1.000 (0.000)	0.000 (0.008)	0.999	0.98	0.747
		2000	1.000 (0.000)	0.000 (0.000)	1.000	1.00	0.749
0.10	heterogeneous	300	0.943 (0.653)	0.050 (0.419)	0.768	0.01	0.638
		600	0.993 (0.250)	0.013 (0.148)	0.930	0.27	0.739
		1000	1.000 (0.000)	0.003 (0.058)	0.986	0.72	0.753
		2000	1.000 (0.000)	0.000 (0.009)	0.999	0.97	0.757

Notes: Point estimates are reported on their original scales. Parenthesized MCSE values for MSE_{η^c}, TPR_η, and FPR_η are multiplied by 100 and reported to three decimal places. The true posterior-score support size is 16. Both $e_B = 0.05$ and $e_B = 0.10$ include equal and heterogeneous concentration trajectories.

Table 6: Stress-condition results over 100 repetitions. Parenthesized scenario triplets give (n, d, q^*).

Scenario	method	$\widehat{q}$	TPR_η	FPR_η	$F_{1,\eta}$	exact	ARI	MSE_{η^c}
Moderately dense $(300, 200, 80)$	Sph. k-means	–	–	–	–	–	0.449	–
	Dense-S	–	–	–	–	–	0.481	4.830
	Dense-F	–	–	–	–	–	0.512	4.393
	M-L	199.50	1.000	0.996	0.572	0.00	0.515	4.360
	E-CGL	15.25	0.154	0.024	0.154	0.00	0.105	6.394
Moderately dense $(1000, 200, 80)$	Sph. k-means	–	–	–	–	–	0.586	–
	Dense-S	–	–	–	–	–	0.600	1.719
	Dense-F	–	–	–	–	–	0.670	0.989
	M-L	199.81	1.000	0.998	0.572	0.00	0.673	0.957
	E-CGL	87.84	0.933	0.110	0.890	0.00	0.674	0.698
Strongly dense $(1000, 200, 160)$	Sph. k-means	–	–	–	–	–	0.589	–
	Dense-S	–	–	–	–	–	0.598	1.716
	Dense-F	–	–	–	–	–	0.669	0.998
	M-L	199.73	1.000	0.993	0.890	0.00	0.671	0.968
	E-CGL	153.96	0.906	0.224	0.924	0.00	0.635	1.547
High-dimensional $(300, 500, 40)$	Sph. k-means	–	–	–	–	–	0.285	–
	Dense-S	–	–	–	–	–	0.405	9.863
	Dense-F	–	–	–	–	–	0.424	9.519
	M-L	499.99	1.000	1.000	0.148	0.00	0.431	9.441
	E-CGL	52.52	0.917	0.034	0.796	0.01	0.759	1.538
Weak beta-min $(1000, 200, 16)$	Sph. k-means	–	–	–	–	–	0.776	–
	Dense-S	–	–	–	–	–	0.768	1.269
	Dense-F	–	–	–	–	–	0.837	0.750
	M-L	199.72	1.000	0.998	0.148	0.00	0.838	0.740
	E-CGL	15.99	0.998	0.000	0.998	0.93	0.869	0.059

Notes: Support metrics use S_η. Dashes indicate that a method does not return a coordinate support. The strongly dense cell is a negative control rather than a setting tailored to sparse selection. Monte Carlo standard errors, test NLL, convergence, endpoint, and runtime summaries are reported in the Supplement.

E-CGL. The sparse-method totals include the paired dense initialization and the method-specific path, refit, and selector; one-time Rcpp compilation was excluded. Because the methods use different path and optimization rules, these values describe computational burden rather than establishing speed superiority. The full seven-method timing panel, iteration counts, failure status, and computing environment are reported in the matched-runtime table in the Supplement.

The primary-grid, sample-size, and stress outputs contain no error rows or duplicate method–replicate keys. Two standard-method cells contained five nonconverged fits among 200 fits: one dense shared-concentration fit and four dense component-specific-concentration fits. Selected E-CGL fits passed the recorded majorization, line-search, and path-endpoint gates. Objective, residual, R/Rcpp-equivalence, and warning-handling diagnostics are reported in the Supplement. These checks are numerical diagnostics and do not establish global optimality.

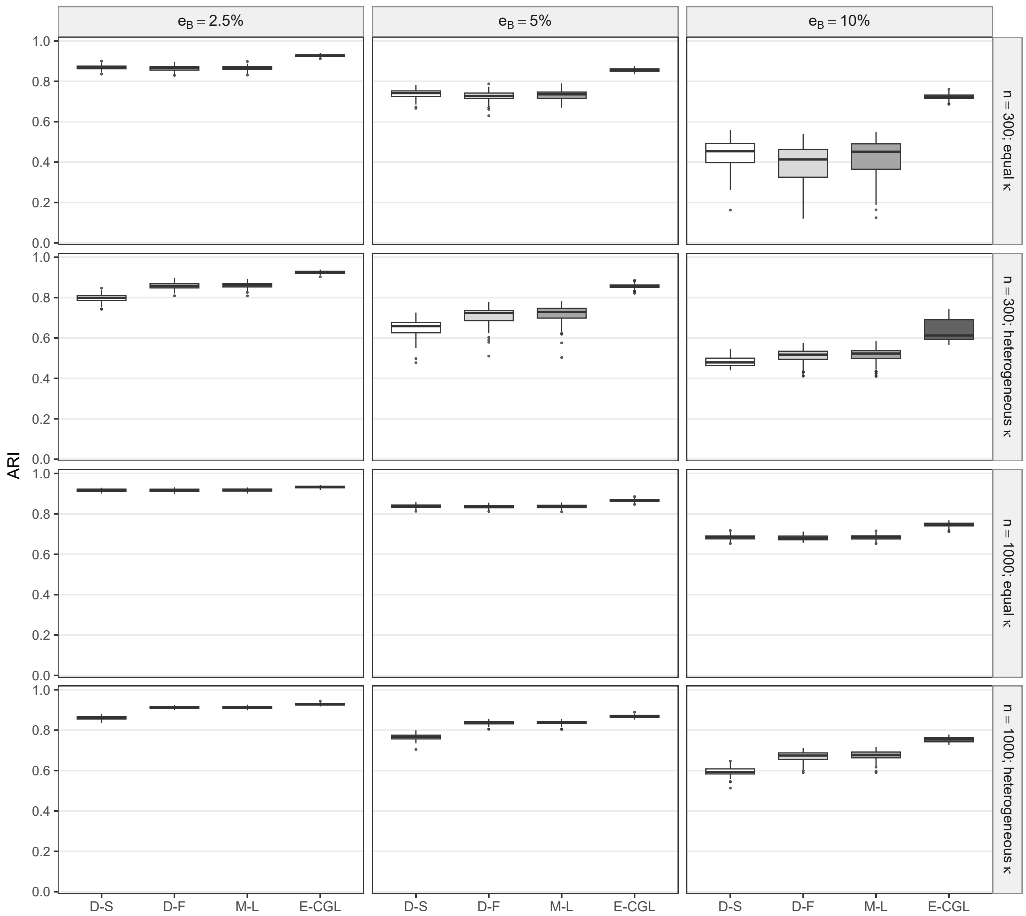

Figure 1: Boxplots of ARI over 100 paired repetitions. Columns index the target oracle Bayes error and rows index sample size and concentration pattern. D-S and D-F denote dense shared- and free-concentration vMF mixtures. M-CGL is omitted because it was prespecified only at $e_B = 0.05$ and targets centered directional rather than posterior-score support; its target-specific results are reported in Table 4.

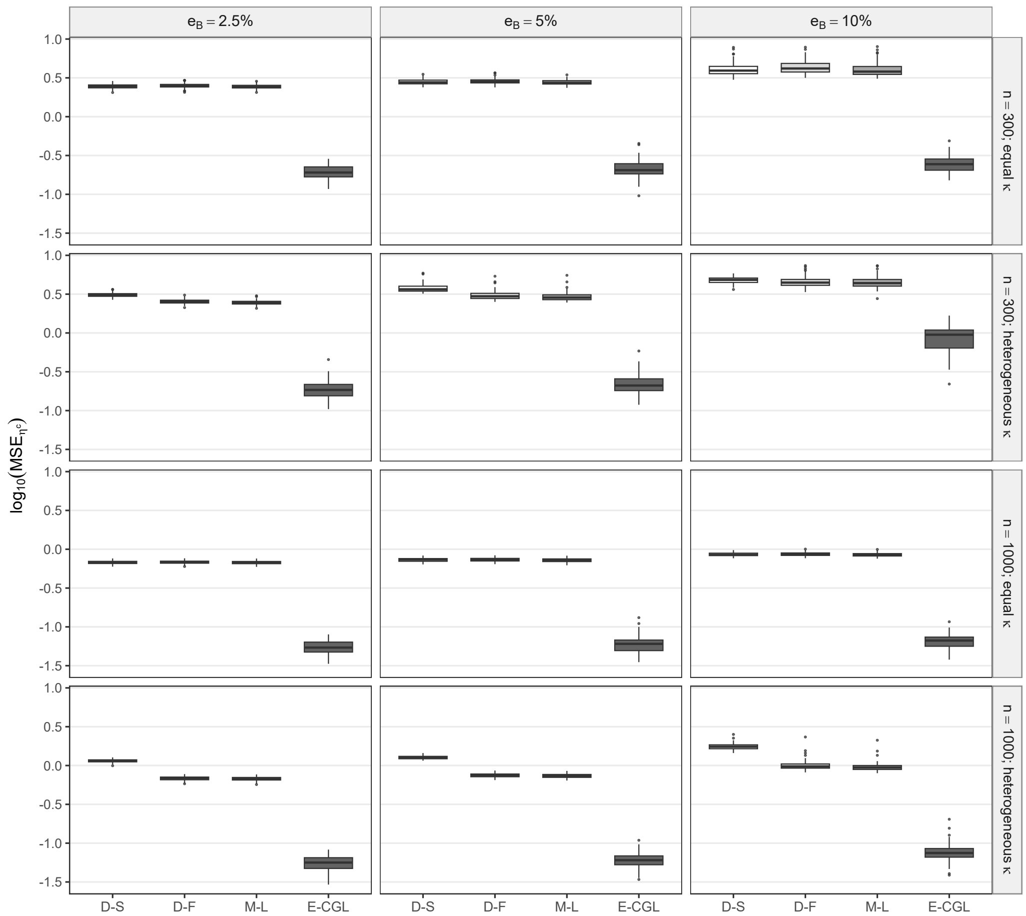

Figure 2: Boxplots of $\log_{10}(\mathrm{MSE}_{\eta^c})$ over 100 paired repetitions. Columns index the target oracle Bayes error and rows index sample size and concentration pattern. D-S and D-F denote dense shared- and free-concentration vMF mixtures. M-CGL is omitted from this display because it was prespecified only at $e_B = 0.05$ and targets centered directional rather than posterior-score support; its target-specific results are reported in Table 4.

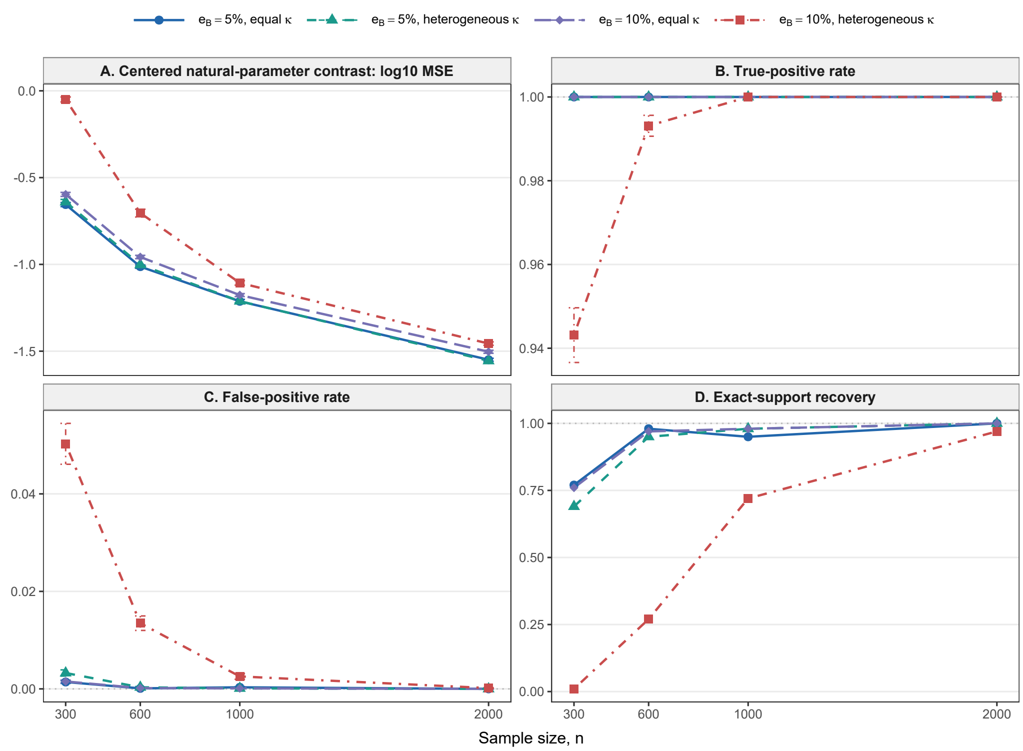

Figure 3: E-CGL sample-size trajectories. Points are Monte Carlo means and error bars are Monte Carlo standard errors over 100 repetitions. Both difficulty levels are shown under equal and heterogeneous concentrations.

5. Real data analysis

5.1. Data and analysis protocol

Classic3 contains abstracts from the CISI, CRAN, and MED document collections and has been used in directional document clustering (Banerjee et al., 2005; Greene and Cunningham, 2006). The raw payload contained $n_0 = 3{,}890$ documents. We removed two exact duplicates and five prespecified near duplicates, leaving $n = 3{,}883$ documents. The cleaned class counts were 1,454, 1,397, and 1,032, respectively.

Documents were encoded with the vocabulary-aligned SPLADE checkpoint, with maximum token length 512 (Formal et al., 2021, 2022). The complete model identifier, revision, and payload checksums are reported in the Supplement. Coordinates occurring in fewer than two documents were removed, the top 2,000 coordinates were ranked by variance over the unlabeled cleaned payload, and each vector was normalized to unit Euclidean norm. The resulting matrix had no zero rows, duplicate transformed rows, duplicate vocabulary entries, or nonfinite values. The model identifier, revision, vocabulary hash, duplicate audit, and payload hash were fixed before model fitting.

The preprocessing rules and split indices were prespecified. E-CGL uses the same single 240-point path and explicit empty-support candidate as in the primary simulations. Adaptive-path diagnostics are reported separately in the Supplement and do not redefine the primary Classic3 estimator.

The benchmark resolution was fixed at $K = 3$ to match the three supplied broad topic labels. The labels were otherwise withheld from coordinate ranking, initialization, parameter estimation, tuning, and support selection; they were used after fitting only to calculate ARI and NMI and to name aligned components. Five prespecified 80/20 splits are retained in the Supplement as stability and held-out diagnostics; they are not pooled with the full-data estimates. After the prespecified duplicate exclusions, each split ranks coordinates using its training documents only. The resulting training-selected coordinate mask is then applied without change to the test documents, after which both matrices are row-normalized.

5.2. Methods and evaluation

The comparison comprised spherical k-means, permutation-tuned sparse k-means (Witten and Tibshirani, 2010), dense vMF mixtures with shared and component-specific concentrations, the Rossi-style M-L prototype selector, M-CGL, and E-CGL. Likelihood-based fits used 30 dense random starts. The M-L path allowed 600 updates. E-CGL used the single zero-augmented 240-point path defined in Section 3, followed by distinct-support refitting, the explicit empty-support candidate, and within-method BIC selection. The same rule was used in each prespecified Classic3 split. E-ACGL is not included in the primary comparison; its standard and augmented path analyses are reported in the Supplement. The sparse vMF methods were selected by their method-specific BIC after support-constrained refitting. Sparse k-means used a label-blind permutation gap criterion with 25 permutations and 10 weight-bound candidates.

Selected q is target-specific: weighted coordinates for sparse k-means, prototype-union coordinates for M-L, centered directional coordinates for M-CGL, and centered natural-parameter coordinates for E-CGL. Accordingly, differences in q do not by themselves rank the estimands. All methods were evaluated by ARI and NMI. Observed log-likelihood and the practical nominal-dimension BIC were additionally reported for vMF models.

5.3. Full-data results and interpretation

The dense free-κ_k mixture attained ARI 0.9760 and NMI 0.9538. E-CGL selected 1,421 of 2,000 coordinates, excluding 28.95% of the coordinates from posterior-score contrasts while attaining ARI 0.9761 and NMI 0.9539. M-CGL gave the same partition with 1,462 directional-contrast coordinates. M-L retained 1,942 prototype coordinates. The E-ACGL sensitivity analysis is reported in the Supplement and is not used to rank the primary E-CGL fit. The permutation-tuned sparse k-means fit selected 985 coordinates but had ARI 0.1374; its selected weight bound was not at the evaluated grid boundary. Independent restarts at that bound reproduced ARI 0.1373 and nearly the same objective value, so this result is reported as performance of the permutation-selected solution under the SPLADE representation rather than as a single-start anomaly or a general statement about sparse k-means.

Table 7: Classic3 full-data results after duplicate removal ($n = 3{,}883$, $d = 2{,}000$, $K = 3$). Component order is aligned to CISI/CRAN/MED only after fitting.

Method	q (q/d)	Cluster sizes	ARI	NMI	ℓ	BIC	$\widehat{\kappa}$
Spherical k-means	2000 (1.000)	1458/1406/1019	0.970	0.944	–	–	–
Sparse k-means	985 (0.492)	553/784/2546	0.137	0.313	–	–	–
Dense vMF shared-kappa	2000 (1.000)	1458/1406/1019	0.970	0.944	18,917,075	-37,784,564	732.2/732.2/732.2
Dense vMF free-kappa	2000 (1.000)	1462/1391/1030	0.976	0.954	18,921,337	-37,793,072	738.5/818.0/615.3
M-L	1942 (0.971)	1462/1405/1016	0.970	0.944	18,914,871	-37,804,965	730.0/730.0/730.0
M-CGL	1462 (0.731)	1456/1392/1035	0.976	0.954	18,919,418	-37,798,125	738.8/816.4/609.9
E-CGL	1421 (0.711)	1456/1392/1035	0.976	0.954	18,919,122	-37,798,211	738.3/815.9/610.4

Selected q refers to the method-specific target: weighted coordinates for sparse k-means, prototype-union coordinates for M-L, centered directional coordinates for M-CGL, and centered natural-parameter coordinates for E-CGL. E-CGL uses the single zero-augmented 240-point standard path and the explicit empty-support candidate, followed by target-preserving refit and within-method BIC selection. Likelihood and BIC are not defined for the two k-means procedures. Method-specific nominal-dimension rules differ, so cross-method BIC differences are descriptive. $\widehat{\kappa}$ is ordered as CISI/CRAN/MED after post-fit component alignment. The E-ACGL sensitivity analysis, including its standard and augmented paths, is reported in the Supplement.

For coordinate j, define

$$R_j = \left(n^{-1} \sum_{i=1}^{n} x_{ij}^2 \right)^{1/2}, \qquad I_{kj}^{(\eta)} = \widehat{c}_{kj} R_j, \qquad I_j^{(\mathrm{com})} = \left| \widehat{\bar{\eta}}_j R_j \right|.$$

The E-CGL concentrations, aligned to CISI, CRAN, and MED only after fitting, were 738.31, 815.88, and 610.41. Panel A of Figure 4 selects the five largest positive root-mean-square-scaled contributions for each matched component and displays their signed centered natural-parameter coefficients. CISI contrasts were led by *library*, *information*, and *librarian*; CRAN by *flow*, *mach*, and *pressure*; and MED by *tumor*, *inhibitor*, and *rat*. Panel B ranks inactive coordinates by $I_j^{(\mathrm{com})}$ and shows those with the largest common root-mean-square-scaled contributions. Their fitted values are equal across components, so their centered contrasts are numerically zero even though the common natural-parameter baseline remains in the fitted density.

5.4. Supplementary stability and scope checks

The Supplement reports the five prespecified Classic3 splits, support Jaccard indices and token-selection frequencies, and the BBCSport contrast analysis. It also retains CSTR as a Rossi-style implementation bridge. In the Classic3 splits, E-CGL had mean conditional support Jaccard 0.933.

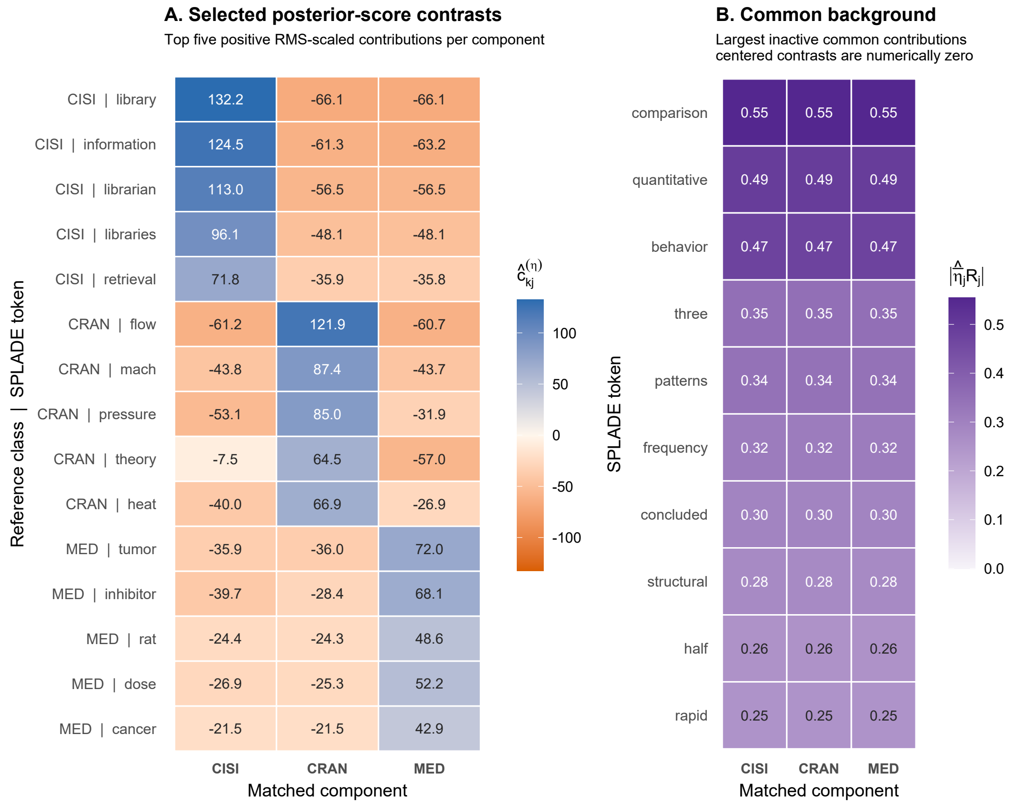

Figure 4: E-CGL coordinate interpretation in Classic3. E-CGL selected 1,421 of 2,000 coordinates. Panel A shows signed centered natural-parameter coefficients for coordinates selected by their positive root-mean-square-scaled contribution to each matched component (blue positive, orange negative). Panel B shows the ten inactive coordinates with the largest absolute common root-mean-square-scaled contribution; their near-identical component values illustrate a retained common background rather than posterior-score heterogeneity. Supplied labels were used only after fitting to align and name components.

BBCSport illustrates a setting in which coordinate reduction incurred held-out density loss, while CSTR is used only to check comparability with the published prototype-sparse analysis. Real data provide no true coordinate support, so support TPR, FPR, precision, and F1 are not reported.

6. Discussion

6.1. Posterior-score contrast support as the estimand

This study defines posterior-score contrast support as the coordinates entering the feature-dependent linear term of pairwise posterior log-score contrasts. E-CGL regularizes centered natural-parameter contrasts and therefore targets posterior-score contrast coordinates. M-L targets prototype support, whereas the M-CGL companion targets centered directional support. Centering leaves the common natural-parameter baseline unpenalized, so a coordinate shared across components is not selected solely because it remains nonzero in every prototype. The coordinate-group penalty is justified by alignment between the regularization unit and this estimand, rather than by uniform empirical superiority over entrywise regularization or prototype-oriented penalties.

6.2. Empirical scope

Across the evaluated $n = 1000$ primary-grid cells, E-CGL selected 16.02–16.47 coordinates on average and attained $F_{1,\eta}$ between 0.986 and 0.999 for a true support size of 16 (Supplementary Tables S1–S2). Clustering ARI decreased with increasing oracle Bayes error, whereas support recovery changed little in these cells; clustering and support difficulty therefore need not vary in parallel in the evaluated settings. The shared-canonical experiment illustrated the practical difference between prototype and posterior-score contrast support. The dense-support experiment, in contrast, documented substantial finite-sample under-selection when decision information was distributed across many weak coordinates, although recovery improved at $n = 1000$ in that design.

The adaptive weighted extension is reported only in the Supplement. Its standard-path results, full augmented rerun, first-positive support audit, and Classic3 analysis are retained there as path-sensitivity evidence. These analyses do not establish an adaptive-weighting advantage and do not redefine the primary E-CGL estimator. The matched penalty-structure ablation also remains supplementary; the primary motivation for E-CGL is alignment between the coordinate-group penalty and the posterior-score support target.

A supplementary target-separation diagnostic evaluated M-CGL against centered directional support and E-CGL against centered natural-parameter support. Both methods recovered their common target under common concentration. In the pure-concentration design, E-CGL recovered the nonempty canonical support with mean target F1 0.990, whereas M-CGL selected the correct empty directional support in 24 of 100 repetitions. In the shared-canonical design, E-CGL recovered the smaller canonical target, whereas M-CGL underselected the larger directional target and did not recover it exactly on the recorded path. These results document path and numerical sensitivity for the directional companion and motivate its diagnostic rather than co-primary role.

The real-data analyses, with prespecified preprocessing and split indices, provided complementary evidence through held-out clustering and support stability rather than recovery of an unknown true support. In the full Classic3 fit, E-CGL excluded 28.95% of the coordinates while closely matching the partition produced by the dense component-specific-concentration model. Across the five prespecified splits it excluded 32.9% on average, with mean ARI and NLL costs of 0.0036 and 0.6705 per document relative to the matched dense vMF fit. Its conditional support Jaccard was 0.933, and the signed token contrasts were coherent across the five splits. In BBCSport, every sparse vMF method had higher held-out NLL than its matched dense model in all five splits. CSTR is retained only as an implementation bridge for the Rossi-style method because its earlier E-series diagnostic used a different selector. These results delimit the intended use of E-CGL rather than establish a uniform ordering of support targets.

The supplementary BBCSport E-CGL result uses the same single standard path as the primary numerical studies; adaptive-path results are reported separately.

6.3. Model selection and computation

The support-recovery claims condition on a fixed number of components. The separate two-stage K diagnostic in Section 3.3.4 is a practical procedure and does not establish consistency of component-number selection. In Classic3, held-out density favored finer component resolutions, whereas partition stability was maximized at the three supplied broad topics. The number of fitted density components and the resolution of an external label system should therefore be reported separately.

Support selection uses the observed likelihood after target-preserving support-constrained refitting (Section 3.3.3). The corresponding BIC nominal dimension is a practical model-selection approximation, rather than an exact effective degrees of freedom result for a nonregular penalized mixture.

Sensitivity analyses showed limited variation among the stored refit candidates under the examined df and EBIC rules, but do not provide a general selection-consistency result.

The fitting algorithm controls accepted generalized-M updates through smooth majorization and backtracking, with a separate step-halving safeguard during support-constrained refitting. Theorem 1 concerns the conditional convex subproblem, and Theorem 2 concerns an idealized sequence under explicit assumptions. The finite-tolerance fitted paths carry no claim of global optimality. Optional Rcpp helpers reproduced the R-only outputs to numerical tolerance and reduced elapsed time in diagnostic benchmarks. This is an implementation result rather than a statistical performance claim.

6.4. Limitations and future directions

E-CGL is intended for settings in which posterior-score contrast support is sparse or compressible. Weak separation, many weak active coordinates, or small samples can produce underselected or empty supports, and mixture fitting remains sensitive to initialization and local optima. In the moderately dense $n = 300$ stress condition, the explicitly added all-common candidate was selected in 81% of E-CGL repetitions, sharply reducing support recovery and clustering performance; this is a limitation of the practical refit BIC under diffuse small-sample signal. The main simulations fix the true K to isolate support recovery, while the separate K experiments provide diagnostics rather than an inferential guarantee. Real-data support cannot be evaluated by TPR or F1 because its true coordinate support is unknown; the evidence instead rests on stability, held-out performance, and substantive interpretation. The Classic3 and BBCSport conclusions are also conditional on the fixed pretrained SPLADE revision and prespecified vocabulary filters. The selected support is coordinate-system dependent: token-level interpretation is meaningful because the representation is aligned with the vocabulary. It does not transfer unchanged to an arbitrarily rotated dense embedding space.

Further work is needed on effective degrees of freedom for centered penalized mixtures, joint uncertainty in K and support, and selection rules that remain stable under diffuse weak signals. The theory fixes K and d; the high-dimensional and $d > n$ results are computational evidence rather than an asymptotic theorem. Adaptive weighting also remains sensitive to pilot contrasts and path coverage. Extensions that combine sparse posterior-score support with explicit dense-background components may also reduce the finite-sample cost observed in dense-support settings.

Centered natural-parameter group regularization defines an estimator of sparse posterior-score contrast support in vMF mixtures. Its target differs from a sparse component prototype: it retains coordinates that change posterior component comparisons while leaving common natural-parameter baselines unpenalized. The simulation and Classic3 analyses support this target-specific interpretation, and the dense-support and supplementary real-data diagnostics identify settings in which sparsification can be costly. E-CGL is therefore positioned as an estimator of posterior-score contrast support for sparse or compressible contrast structures, not as a uniformly superior replacement for dense or prototype-oriented mixture methods.

Appendix A. Proofs of theoretical results

Proof of Lemma 1. The closed simplex and the product of the K closed Euclidean balls of radius $\kappa_{\max}$ are compact. Each vMF component density is positive and continuous on this set, so the observed log-likelihood is finite and continuous. The centered group penalty is also continuous. The criterion therefore attains its maximum by the extreme-value theorem. □

Proof of Proposition 1. Taking the average of $v = \bar{v}\mathbf{1}_K + c$ and using $\mathbf{1}_K^\top c = 0$ determines $\bar{v}$ and hence c uniquely. The three zero-contrast statements are immediate. Finally, $\sum_{k<\ell}(v_k - v_\ell)^2 = K\sum_k(v_k - \bar{v})^2$, which gives the stated identity. □

Proof of Proposition 2. Because $\kappa = \bar{\kappa}\mathbf{1}_K + h$ and $M = \mathbf{1}_K\bar{\mu}^\top + U$,

$$E = \operatorname{diag}(\kappa)M = \bar{\kappa}\mathbf{1}_K\bar{\mu}^\top + \bar{\kappa}U + h\bar{\mu}^\top + \operatorname{diag}(h)U.$$

Apply H_K, using $H_K\mathbf{1}_K = 0$, $H_KU = U$, and $H_Kh = h$. □

Proof of Proposition 3. Because $P\mathbf{1}_K = \mathbf{1}_K$, P commutes with J_K and H_K. Permutation matrices are orthogonal and therefore preserve the Euclidean norm. Applying Proposition 4 gives equivariance of the proximal map. □

Proof of Proposition 4. The spaces spanned by $\mathbf{1}_K$ and orthogonal to $\mathbf{1}_K$ are orthogonal. The penalty is zero on the first space and equals the Euclidean norm on the second. The minimization therefore preserves J_Kv and applies the standard group soft-thresholding map to H_Kv. □

Proof of Proposition 5. The vMF log-partition function has Hessian equal to the covariance matrix of a unit-norm random vector and is therefore positive semidefinite with operator norm at most one. The Hessian of the kth block of f_t consequently has operator norm at most N_k. The descent result follows by combining the smooth majorization inequality with optimality of the proximal update. $\square$

Proof of Theorem 1. For finite natural parameters, the vMF log-partition function is strictly convex. Its Hessian is the covariance matrix of a vMF random vector, and the vMF density is positive on the full sphere; hence $\mathrm{Var}(a^\top X) > 0$ for every nonzero $a \in \mathbb{R}^d$. Since $N_k > 0$, the sum of its K blocks is strictly convex. The standard large-argument expansion of the modified Bessel function gives

$$-\log C_d(\kappa) = \kappa - \frac{d-1}{2}\log\kappa + O(1), \qquad \kappa \longrightarrow \infty.$$

Consequently, as $\|\eta_k\|_2 \to \infty$,

$$N_k\{-\log C_d(\|\eta_k\|_2)\} - r_k^\top \eta_k \geq (N_k - \|r_k\|_2)\|\eta_k\|_2 - \frac{N_k(d-1)}{2}\log\|\eta_k\|_2 + O(1).$$

The positive linear coefficient implied by $\|r_k\|_2 < N_k$ dominates the logarithmic term. Thus f_t is coercive. Adding the finite convex group penalty preserves strict convexity and coercivity, so F_t has a unique minimizer. If that minimizer is interior to the concentration bound, it is also the unique bounded-subproblem solution. The gradient of f_t is globally Lipschitz with constant no larger than $\max_k N_k$. Hence the initial trial step $1/\max_k N_k$ satisfies the exact majorization inequality and is bounded away from zero. Standard proximal- gradient theory yields convergence and the $O(m^{-1})$ objective rate (Beck and Teboulle, 2009). The strict inequality $\|r_k\|_2 < N_k$ excludes the degenerate case in which all observations receiving positive responsibility from a component have one common direction. The finite cap still gives existence in that case, but this theorem does not apply. $\square$

Proof of Proposition 6. The standard EM lower-bound identity expresses the observed log-likelihood difference as the auxiliary-function difference plus a nonnegative Kullback–Leibler divergence term. The penalty depends only on the parameters and can be subtracted from both sides, yielding the penalized result. $\square$

Proof of Theorem 2. Let $\Theta^{(t_j)} \to \bar{\Theta}$ be any convergent subsequence, which exists by compactness. Asymptotic regularity also gives $\Theta^{(t_j+1)} \to \bar{\Theta}$. After passage to a further subsequence if needed, boundedness of the step sizes gives $s_{t_j} \to \bar{s}$ for some $\bar{s} \in [\underline{s}, \overline{s}]$. Exact posterior responsibilities are continuous on the assumed compact interior set, so $\tau^{(t_j)} \to \tau(\bar{\Theta})$.

The gradient $\nabla f_\tau(E)$ is jointly continuous in (E, τ) on this set. For s bounded away from zero, the scaled proximal map is jointly continuous in its input and in s. Moreover, $s_{t_j} \mathcal{G}_{s_{t_j}}(E^{(t_j+1)}; \tau^{(t_j)}) \to 0$. The vanishing natural-parameter residual therefore implies

$$\bar{E} = \mathrm{prox}_{\bar{s}\lambda_\eta P}\{\bar{E} - \bar{s}\nabla f_{\tau(\bar{\Theta})}(\bar{E})\}.$$

The proximal fixed-point characterization yields

$$0 \in \nabla f_{\tau(\bar{\Theta})}(\bar{E}) + \lambda_\eta \partial P(\bar{E}).$$

At exact posterior responsibilities, the Fisher–EM score identity gives $\nabla f_{\tau(\bar{\Theta})}(\bar{E}) = -\nabla_E \ell(\bar{\Theta})$, proving the natural-parameter KKT inclusion. For the mixing block,

$$\pi^{(t_j+1)} - \frac{N(\tau^{(t_j)})}{n} \longrightarrow 0,$$

while $\pi^{(t_j+1)} \to \bar{\pi}$ and $N(\tau^{(t_j)}) \to N\{\tau(\bar{\Theta})\}$. Hence $\bar{\pi}_k = N_k\{\tau(\bar{\Theta})\}/n$. Because the limiting mixing proportions are strictly positive,

$$\frac{\partial \ell(\bar{\Theta})}{\partial \pi_k} = \frac{N_k\{\tau(\bar{\Theta})\}}{\bar{\pi}_k} = n$$

for every k, which is the interior simplex KKT condition with $\nu = n$. The objective, residuals, and KKT system are equivariant under any fixed component permutation. The argument does not apply to boundary or finite-tolerance cases excluded by the assumptions (Green, 1990; Wu, 1983; Tseng, 2004). □

Data and code availability

The source corpora must be obtained from their original providers under the applicable terms. Derived document-level SPLADE matrices are not redistributed because redistribution permission for the source corpora has not been established. A redistribution-safe reproducibility archive containing the reviewed implementation, tests, preprocessing specifications, configuration files, random seeds and split records, exclusion records, vocabulary and payload checksums, compact numerical summaries, and result-generation scripts is available from the corresponding author on reasonable request. Large raw, path, cache, and private project files are not part of that archive.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

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